SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 1 Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I K Giridhar (192425160) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K Giridhar

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary (5)**	**Level 3 – Proficient (4)**	**Level 2 – Developing (2-3)**	**Level 1 – Beginning (0-1)**
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Error Analysis: Neural Network Performance: (Training MSE = 0.85, Testing MSE = 5.20)

Problem Statement:

A neural network is used to predict student scores. During evaluation, the following Mean Squared Error (MSE) values are obtained:

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

Analyze the possible reasons for the difference and suggest suitable corrective measures.

Understanding Mean Squared Error (MSE)

Mean Squared Error (MSE) measures the average squared difference between the actual values and the predicted values.

- **Lower MSE** indicates better prediction accuracy.
- **Higher MSE** indicates larger prediction errors.

Formula:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y_i})^2$$

where:

- y_i= Actual value
- $\hat{y_i}$= Predicted value
- n= Number of samples

Error Analysis:

Metric	Value	Interpretation
Training MSE	0.85	Very low error, indicating excellent performance on training data.
Testing MSE	5.20	High error, indicating poor prediction on unseen data.
Difference	4.35	Large gap between training and testing performance.

Observation:

The neural network performs very well on the training dataset but poorly on the testing dataset. This is a classic symptom of **overfitting**, where the model memorizes the training data instead of learning general patterns.

Possible Reasons:

Overfitting:

The model learns the training data too closely, including noise. As a result, it performs poorly on new testing data.

Complex Neural Network:

Using too many hidden layers or neurons makes the model overly complex. This increases the chance of overfitting.

Small Training Dataset:
A limited amount of training data does not provide enough examples for learning. This reduces the model's ability to generalize.

No Regularization:
Without techniques like Dropout or L2 regularization, the model can easily memorize the training data instead of learning patterns.

Corrective Measures

Reduce Model Complexity:
Use fewer hidden layers or neurons to make the model simpler. This helps improve generalization.

Use Dropout or L2 Regularization:
These techniques prevent overfitting by reducing the dependence on specific neurons and controlling large weights.

Increase Training Data:
Training the model with more data helps it learn better patterns and improves testing performance.

Apply Early Stopping:
Stop training when the validation error starts increasing. This prevents the model from overfitting the training data.

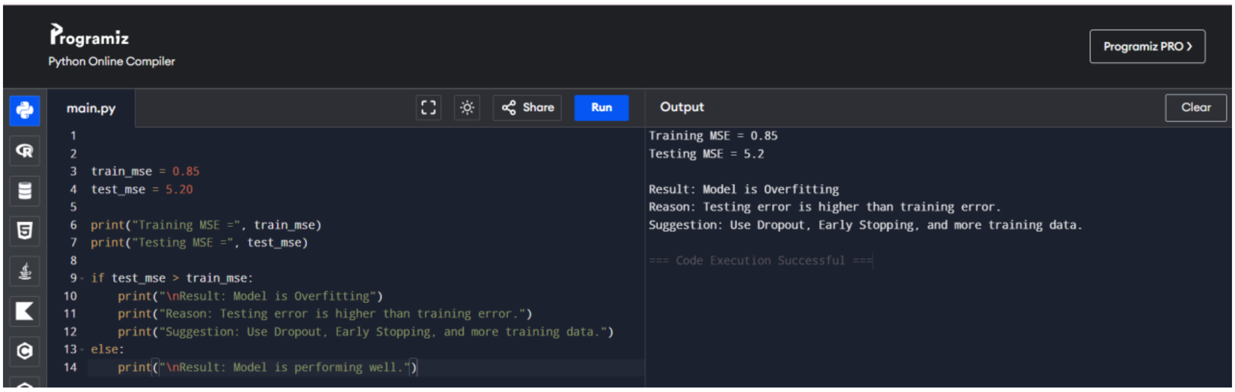
Use Cross Validation:
Evaluate the model using different data splits to obtain more reliable and accurate performance results.

Python program:

```
train_mse = 0.85
test_mse = 5.20

print("Training MSE =", train_mse)
print("Testing MSE =", test_mse)

if test_mse > train_mse:
   print("\nResult: Model is Overfitting")
   print("Reason: Testing error is higher than training error.")
   print("Suggestion: Use Dropout, Early Stopping, and more training data.")
else:
   print("\nResult: Model is performing well.")
```



Result:

The neural network achieved a **Training MSE of 0.85** and a **Testing MSE of 5.20**. Since the testing error is much higher than the training error, the model is **overfitting**. Applying techniques such as **Dropout, Early Stopping, regularization, and increasing the training data** can improve the model's performance on unseen data.